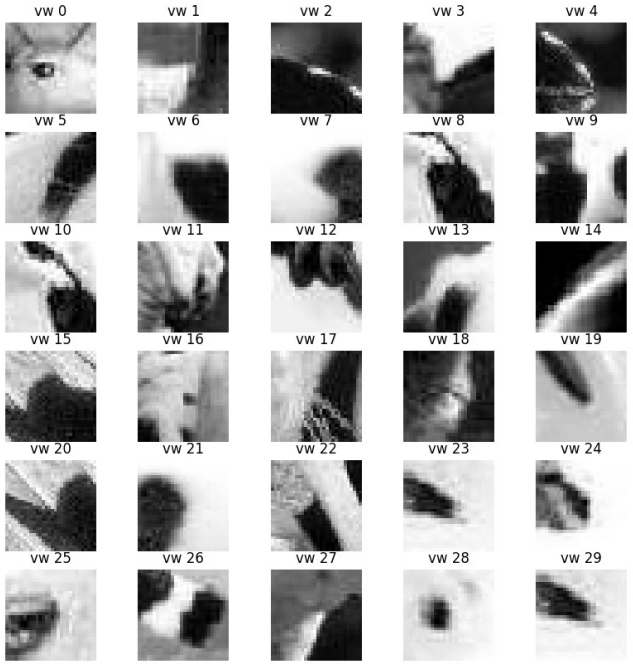
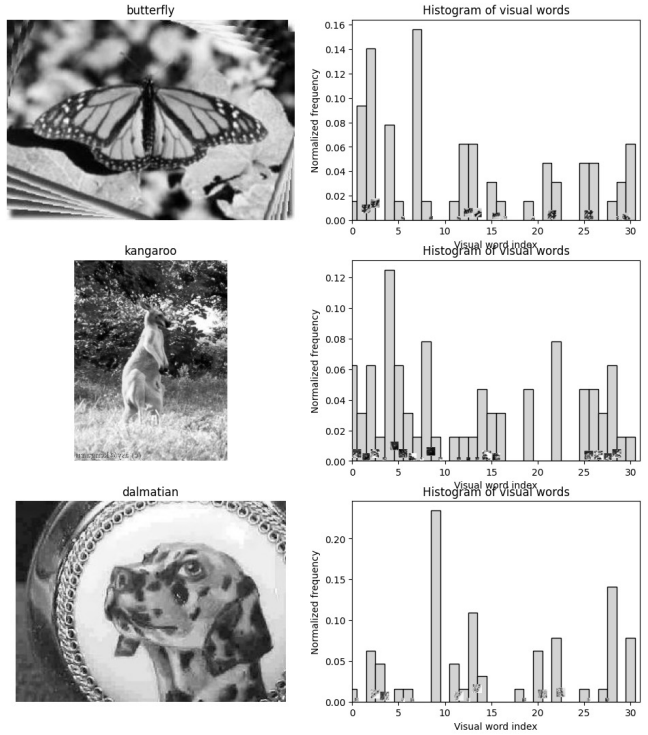
# Instructions for Set 7

- Codebook construction
- Histogram construction
- Classification



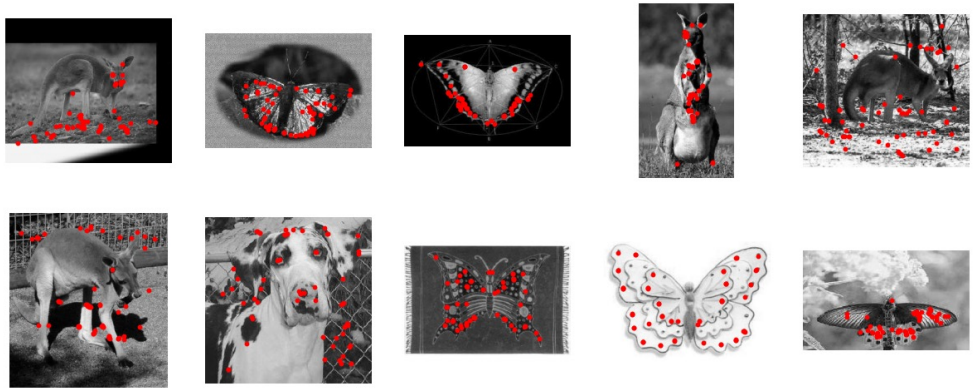




p(kangaroo) = 0.64

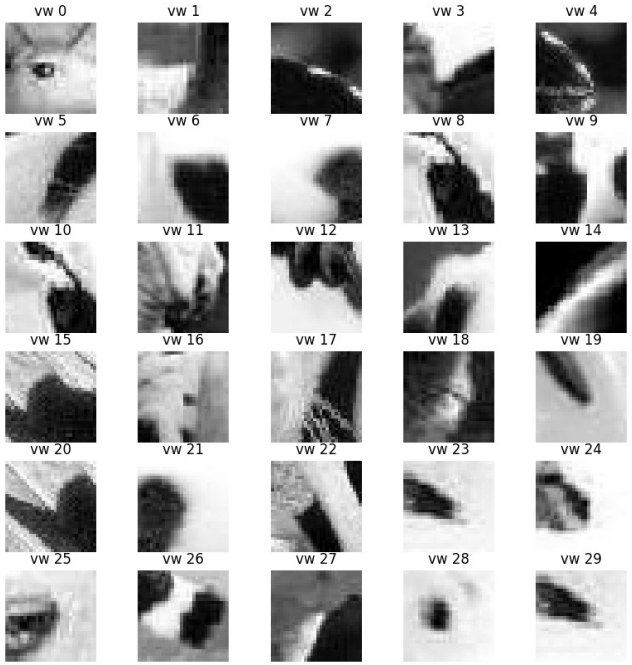


# Set 7.1 Codebook construction

- Cluster descriptors using K-Means (K = 10*num_classes)
- Define and visualize visual words:
  - each cluster centre becomes the visual word

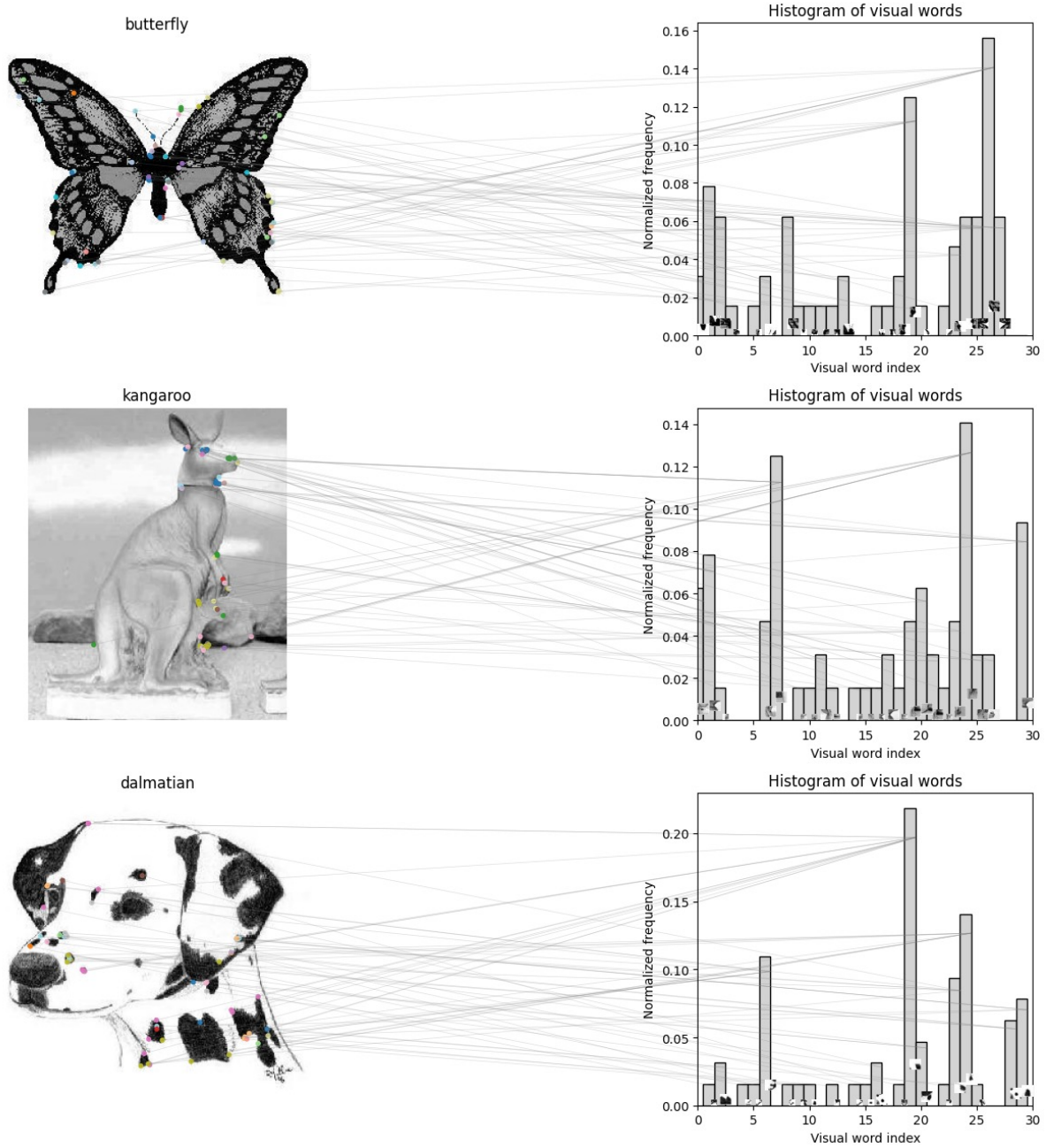


```
from sklearn.cluster import MiniBatchKMeans
descriptors = feature_descriptors[0]
for descriptor in feature_descriptors[1:]:
    descriptors=np.vstack((descriptors,descriptor))
print(descriptors.shape)
kmeans = MiniBatchKMeans(n_clusters=ncl, random_state=0).fit(descriptors)
print(kmeans)
```



# Set 7.2 Histogram construction

- Assign features to words:
  - map each feature to its closest centroid
- Build a frequency histogram
  - k-dimensional histogram
- Visualize the histogram (for test images only)



# Set 7.3 Classification

- Given a new image
  - Visualize the histogram (from 7.2)
  - After clicking ”classify”:
    - run the trained classifier
    - show the top probability

p(kangaroo) = 0.64

